Asthma and COPD: Speaker Notes

English for Medicine II. Group presentation, three presenters, about 8 minutes each (allowed: 7 to 10 minutes). Times do not include questions.

How to use these notes

- The same script is in each slide's notes, so you can read it in PowerPoint's Presenter View.
- `click` means: press the clicker once. Every `click` matches one animation on the slide.
- Stressed syllables are in CAPITALS: AZ-muh = **AZ**-muh. Practice the words under *Say carefully* aloud.
- Spelling is American English (edema, hypoxemia, fiber).
- To add your names, use Find and Replace for "Presenter 1", "Presenter 2" and "Presenter 3" (Home > Replace), in the slides and in this file.

Timing

Presenter	Slides	Part	Words	100 wpm	110 wpm	120 wpm
Presenter 1	1 to 9	Opening, airway obstruction and asthma	878	9:10	8:22	7:42
Presenter 2	10 to 17	COPD and spirometry	881	9:12	8:24	7:44
Presenter 3	18 to 26	Drugs, treatment, quiz and closing	895	9:23	8:34	7:54
Total	1 to 26		**2654**	27:44	**25:20**	23:19

Most people speak at 100 to 120 words per minute in a presentation, so every part fits in 7 to 10 minutes. Rehearse with a timer.

Presenter 1: Opening, airway obstruction and asthma (slides 1 to 9)

Total: about 8:22 at 110 words per minute (878 words)

Slide 1: Asthma and COPD

What to say: Please take a deep breath in… and now breathe out. Easy, right? But for hundreds of millions of people, breathing out is the hard part. Hello, everyone. We are Presenter 1, Presenter 2 and Presenter 3. Today we will talk about asthma and COPD, the two most common obstructive lung diseases, and about the drugs that open the airways again.

Say carefully: asthma = AZ-muh; **COPD** = see-oh-pee-DEE; **obstructive** = ub-STRUK-tiv

Time: about 35 seconds, no clicks

Slide 2: Two Patients

What to say: Let me introduce two patients. click This is Sara. She is fourteen. She wheezes and coughs after football, and she has eczema. Between attacks, she feels completely normal. Sara has asthma, like about three hundred and sixty million people around the world. click This is Mr. Brown. He is sixty-two, and he has smoked a pack of cigarettes a day for forty years. He coughs up sputum every morning, and the stairs make him more and more breathless. Mr. Brown has COPD, the third leading cause of death in the world. click Both patients have trouble breathing out. But Sara's obstruction comes and goes, while Mr. Brown's stays. Why? By the end of our talk, you will be able to explain it.

Say carefully: wheeze = WEEZ; **eczema** = EK-zuh-muh; **sputum** = SPYOO-tum; **breathless** = BRETH-lus

Time: about 70 seconds, 3 clicks

Slide 3: Outline

What to say: To answer this question, our talk has three parts. First, I will explain what airway obstruction means and what happens inside the airways in asthma. Then Presenter 2 will explain COPD and show how a simple breathing test tells the two diseases apart. Finally, Presenter 3 will explain how the main drugs work. At the end, we have a short quiz, and we will come back to Sara and Mr. Brown.

Say carefully: obstruction = ub-STRUK-shun

Time: about 40 seconds, no clicks

Slide 4: Airway Obstruction and Asthma

What to say: Let's start with part one: airway obstruction and asthma.

Time: about 5 seconds, no clicks

Slide 5: Airway Narrowing

What to say: First, what does obstruction mean? You know from physiology that airflow depends on pressure and resistance, and resistance depends on the radius of the airway to the fourth power. So if the radius falls by half, the resistance rises sixteen times. Also remember that breathing out is normally passive: the lungs simply recoil, like a balloon. In obstructive disease, the air cannot leave the lungs fast enough, so some of it stays inside. We call this air trapping. There are three ways to narrow an airway. click The first is bronchoconstriction: the smooth muscle contracts. click The second is inflammation: the wall becomes thick and swollen, and mucus fills the lumen. click The third is collapse. The alveoli around a small airway hold it open, like ropes holding up a tent. If these walls are destroyed, the airway collapses when the patient breathes out. click Asthma mainly uses the first two ways, and they can be reversed. COPD mainly uses the last two, and they are persistent.

Say carefully: Poiseuille = pwah-ZOY; **resistance** = rih-ZIS-tuns; **radius** = RAY-dee-us; **recoil (verb)** = rih-KOYL; **bronchoconstriction** = BRONG-koh-kun-STRIK-shun; **lumen** = LOO-men; **alveoli** = al-VEE-uh-lye; **persistent** = per-SIS-tunt

Time: about 95 seconds, 4 clicks

Slide 6: What Is Asthma?

What to say: Now let's look at asthma. click Asthma is a chronic inflammatory disease of the airways. click The inflammation makes the airways hyperresponsive. This means they overreact: a small trigger, like cold air or dust, causes strong bronchoconstriction. click So the obstruction is variable: it comes and goes, and it is usually reversible. click The main symptoms are wheeze, cough, chest tightness and shortness of breath. When air rushes through narrow airways, the walls vibrate, and we hear a wheeze. click The strongest risk factor is atopy, a genetic tendency to make IgE antibodies against harmless things, such as dust mites or pollen. Sara's eczema is a clue that she is atopic. So how does this inflammation start?

Say carefully: inflammatory = in-FLAM-uh-tor-ee; **hyperresponsive** = HY-per-rih-SPON-siv; **dyspnea** = DISP-nee-uh; **atopy** = AT-uh-pee; **atopic** = ay-TOP-ik; **IgE** = eye-jee-EE

Time: about 65 seconds, 5 clicks

Slide 7: Pathophysiology of Asthma: Sensitization

What to say: Allergic asthma is a type one hypersensitivity reaction, and it begins with sensitization. click An allergen, for example a dust mite protein, enters the airway. A dendritic cell picks it up and presents it to a T helper cell. In atopic people, this cell becomes a Th2 cell. click The Th2 cell releases three important interleukins. click Interleukin four tells B cells to make IgE. Interleukin five calls eosinophils. And interleukin thirteen increases mucus and makes the airways hyperresponsive. The IgE then binds to mast cells in the airway wall. Now the mast cells are armed. Nothing happens yet, but the next time the allergen arrives, they are ready.

Say carefully: hypersensitivity = HY-per-sen-sih-TIV-ih-tee; **sensitization** = SEN-sih-tih-ZAY-shun; **allergen** = AL-er-jen; **dendritic** = den-DRIT-ik; **Th2** = tee-aitch-TOO; **interleukin** = IN-ter-LOO-kin; **eosinophil** = ee-uh-SIN-uh-fil

Time: about 60 seconds, 3 clicks

Slide 8: Early and Late Phase Reaction

What to say: When Sara meets the allergen again, her airways narrow in two waves. This graph shows her FEV1 after she breathes in the allergen. click The first wave is the early phase. Within minutes, the allergen cross-links the IgE on the mast cells. They release histamine and leukotrienes, the smooth muscle contracts, and the FEV1 falls. After an hour or two, it recovers. click But in many patients, a second wave comes four to eight hours later. This is the late phase. Eosinophils and Th2 cells arrive and cause inflammation, edema and mucus. This matters for treatment: a bronchodilator relieves the early phase, but we need a corticosteroid to prevent the late phase. click And over many years, repeated inflammation remodels the airway, so some obstruction can become permanent.

Say carefully: FEV1 = eff-ee-vee-WUN; **histamine** = HIS-tuh-meen; **leukotriene** = loo-kuh-TRY-een; **edema** = ih-DEE-muh; **bronchodilator** = BRONG-koh-dy-LAY-ter; **corticosteroid** = KOR-tih-koh-STEER-oyd; **remodel** = ree-MOD-ul

Time: about 70 seconds, 3 clicks

Slide 9: Why Do Symptoms Come and Go?

What to say: Finally, why do Sara's symptoms come and go? click The first trigger is exercise. When she runs, she breathes fast through her mouth. This cools and dries the airways, and the mast cells release their mediators. click The second is the night. In the early morning, vagal tone is higher, and cortisol and adrenaline are lower, so many patients wake up coughing. click The third is viral infections, the most common cause of severe attacks. click And of course, allergens such as pollen, dust mites and cat dander. click To sum up: asthma is Type 2 inflammation plus overreactive smooth muscle, and the obstruction comes and goes. Now Presenter 2 will show you a very different disease: COPD.

Say carefully: mediator = MEE-dee-ay-ter; **vagal** = VAY-gul; **cortisol** = KOR-tih-sol; **adrenaline** = uh-DREN-uh-lin; **viral** = VY-rul; **dander** = DAN-der

Time: about 65 seconds, 5 clicks

Questions you might get (Presenter 1)

1. **Q:** Is all asthma allergic?
   A: No. Allergic, or Type 2, asthma is the most common type, especially in children. But some adults have non-allergic asthma. Their inflammation is driven by other cells, such as neutrophils, and they often respond less well to steroids.
2. **Q:** Why does aspirin trigger asthma in some patients?
   A: Aspirin blocks cyclooxygenase. Arachidonic acid is then shifted to the lipoxygenase pathway, so the body makes more leukotrienes, which contract the airways. Many of these patients also have nasal polyps. A leukotriene blocker such as montelukast can help.
3. **Q:** Can asthma turn into COPD?
   A: Long-standing asthma can remodel the airways, so some obstruction becomes fixed, and childhood asthma is a risk factor for COPD. Some patients have features of both diseases, and they are treated for both.

Presenter 2: COPD and spirometry (slides 10 to 17)

Total: about 8:24 at 110 words per minute (881 words)

Slide 10: COPD

What to say: Thank you, Presenter 1. In part two, I will talk about COPD, and how we can tell it apart from asthma.

Time: about 10 seconds, no clicks

Slide 11: What Is COPD?

What to say: COPD stands for chronic obstructive pulmonary disease. It is airflow obstruction that is persistent and often progressive. It comes from damage to the airways, to the alveoli, or usually to both. click The airway part is chronic bronchitis. It is defined by symptoms: a cough with sputum for at least three months a year, for two years in a row. click The alveolar part is emphysema. It is defined by structure: the alveolar walls are destroyed, so the air spaces become larger. Most patients, like Mr. Brown, have both.

Say carefully: chronic = KRON-ik; **pulmonary** = PUL-muh-nair-ee; **progressive** = pruh-GRES-iv; **bronchitis** = brong-KY-tis; **emphysema** = em-fuh-SEE-muh; **alveolar** = al-VEE-uh-ler

Time: about 50 seconds, 2 clicks

Slide 12: Causes of COPD

What to say: What causes this damage? click The main cause is tobacco smoke, and we measure the dose in pack-years. Mr. Brown has forty. click But in many low- and middle-income countries, smoke from cooking fires is also a major cause. Look at this kitchen in Nepal: the black walls show the smoke that the family breathes every day. click Dust and fumes at work and outdoor air pollution also play a role. click And there is a rare genetic cause, alpha-1 antitrypsin deficiency. You will see why it matters on the next slide.

Say carefully: pack-year = PAK-yeer; **fumes** = FYOOMZ; **alpha-1 antitrypsin** = AL-fuh WUN an-tee-TRIP-sin; **deficiency** = dih-FISH-un-see

Time: about 50 seconds, 4 clicks

Slide 13: Emphysema: Protease-Antiprotease Imbalance

What to say: So how does smoke destroy the alveoli? The key idea is a balance between proteases and antiproteases. click Smoke brings macrophages and neutrophils into the lung. These cells release proteases, such as elastase. Elastase digests elastin, the elastic fiber in the alveolar wall. click Normally, alpha-1 antitrypsin, a protein made by the liver, blocks elastase. But oxidants in smoke inactivate it. So elastase wins, and the walls are destroyed. click The pattern can show the cause. In smokers, the damage starts in the center of the acinus, mainly in the upper lobes. This is centriacinar emphysema. In alpha-1 antitrypsin deficiency, a genetic disease, the whole acinus is damaged, mainly in the lower lobes, and often in younger patients. This is panacinar emphysema.

Say carefully: protease = PROH-tee-ayz; **macrophage** = MAK-ruh-fayj; **neutrophil** = NOO-truh-fil; **elastase** = ih-LAS-tayz; **elastin** = ih-LAS-tin; **oxidant** = OK-sih-dunt; **acinus** = AS-ih-nus; **centriacinar** = SEN-tree-AS-ih-ner; **panacinar** = pan-AS-ih-ner

Time: about 70 seconds, 3 clicks

Slide 14: Loss of Elastic Recoil and Air Trapping

What to say: Now, what does this destruction do to breathing? click Without elastin, the lung loses its elastic recoil, and its compliance goes up: it fills easily, but it does not spring back. click The small airways also lose the alveolar walls that held them open, like ropes holding up a tent. So when Mr. Brown breathes out, the pressure around these airways squeezes them shut. click Air is trapped behind them, and the lungs become hyperinflated. Look at the chart: the residual volume, the FRC and the total lung capacity are all higher than normal. click This explains two signs: the diaphragm becomes flat, and the chest becomes round, like a barrel. Breathing with a flat diaphragm costs a lot of energy, so many patients lose weight. click It also explains a trick that patients learn: pursed-lip breathing. Breathing out through tight lips keeps the pressure inside the airways higher, so they stay open longer.

Say carefully: recoil (noun) = REE-koyl; **compliance** = kum-PLY-uns; **hyperinflated** = HY-per-in-FLAY-tid; **residual** = rih-ZID-yoo-ul; **diaphragm** = DY-uh-fram; **pursed-lip** = PURST-lip

Time: about 85 seconds, 5 clicks

Slide 15: Chronic Bronchitis

What to say: Now the airway side: chronic bronchitis. click Smoke irritates the airways. The mucus glands become larger, and there are more goblet cells, so the airways make too much mucus. click At the same time, smoke damages the cilia, so the mucus is not cleared. It collects during the night, and this explains Mr. Brown's cough with sputum every morning. click In the small airways, inflammation and fibrosis thicken the walls, and mucus plugs them. This is the second way to narrow an airway, but in COPD it does not reverse. The trapped mucus also helps bacteria grow, so infections often cause flare-ups.

Say carefully: goblet cell = GOB-lit sel; **cilia** = SIL-ee-uh; **fibrosis** = fy-BROH-sis; **flare-up** = FLAIR-up

Time: about 55 seconds, 3 clicks

Slide 16: Hypoxemia and Cor Pulmonale

What to say: When the airways and alveoli are damaged, ventilation and perfusion no longer match. Some alveoli get blood but very little air. So the blood leaves the lungs with less oxygen, which is hypoxemia, and in severe disease with more carbon dioxide, which is hypercapnia. click You know that pulmonary arteries constrict when the oxygen is low. This is hypoxic vasoconstriction. In a small area, it is helpful. But in COPD the hypoxia is everywhere, so the arteries constrict everywhere, and emphysema also destroys capillaries. click The result is pulmonary hypertension. click The right ventricle must pump against this high pressure, so it hypertrophies, and finally it fails. Right heart failure caused by lung disease is called cor pulmonale. The patient develops ankle edema and a raised jugular venous pressure. click Low oxygen also makes the kidneys release more erythropoietin, which causes polycythemia.

Say carefully: ventilation = ven-tih-LAY-shun; **perfusion** = per-FYOO-zhun; **hypoxemia** = hy-pok-SEE-mee-uh; **hypercapnia** = hy-per-KAP-nee-uh; **vasoconstriction** = VAY-zoh-kun-STRIK-shun; **hypertension** = HY-per-TEN-shun; **ventricle** = VEN-trih-kul; **hypertrophies** = hy-PER-truh-feez; **cor pulmonale** = KOR pul-muh-NAH-lee; **jugular** = JUG-yuh-ler; **erythropoietin** = ih-RITH-roh-POY-uh-tin; **polycythemia** = POL-ee-sy-THEE-mee-uh

Time: about 80 seconds, 4 clicks

Slide 17: Spirometry: Asthma vs COPD

What to say: So how do we tell the two diseases apart? The key test is spirometry. The patient breathes in fully, then breathes out as hard and as fast as possible. FEV1 is the volume breathed out in the first second, and FVC is the total volume. In obstruction, the FEV1 falls more than the FVC, so the FEV1/FVC ratio drops below zero point seven. Look at the flow-volume loop: the obstructed loop is scooped, because flow falls quickly as the airways close, and it is shifted to higher volumes because of air trapping. **click** Then we give a bronchodilator and test again. In asthma, the FEV1 rises by at least twelve percent and two hundred milliliters, so the obstruction is reversible. Between attacks, the test can even be normal. In COPD, the ratio stays below zero point seven, so the obstruction is persistent. **click** One more test helps: the DLCO, the diffusing capacity for carbon monoxide. It is low in emphysema, because the alveolar walls and capillaries are gone, but normal in asthma. Now Presenter 3 will explain how drugs open these airways.

Say carefully: spirometry = spy-ROM-uh-tree; **FVC** = eff-vee-SEE; **ratio** = RAY-shee-oh; **milliliters** = MIL-ih-lee-terz; **DLCO** = dee-el-see-OH; **diffusing capacity** = dih-FYOO-zing kuh-PAS-ih-tee

Time: about 100 seconds, 2 clicks

Questions you might get (Presenter 2)

1. **Q:** Why doesn’t every smoker develop COPD?
   A: Susceptibility differs. Genes, such as alpha-1 antitrypsin variants, lung growth in early life and other exposures all play a role. In Fletcher and Peto’s study, only some smokers were “susceptible” and lost lung function quickly.
2. **Q:** Why can alpha-1 antitrypsin deficiency also damage the liver?
   A: In the severe form, the abnormal protein misfolds and gets stuck inside the liver cells that make it. This can cause cirrhosis. At the same time, too little of the protein reaches the lungs.
3. **Q:** Why is DLCO normal in chronic bronchitis but low in emphysema?
   A: DLCO measures gas transfer across the alveolar-capillary membrane. In emphysema, the alveolar walls and their capillaries are destroyed, so the surface area is smaller. Chronic bronchitis is a disease of the airways, so the membrane is intact.

Presenter 3: Drugs, treatment, quiz and closing (slides 18 to 26)

Total: about 8:34 at 110 words per minute (895 words)

Slide 18: Drugs and Treatment

What to say: Thank you, Presenter 2. In the last part, I will explain how the main drugs work. You will see that each drug targets one of the mechanisms we have just described.

Time: about 15 seconds, no clicks

Slide 19: Bronchodilators: How They Work

What to say: Let's start with the airway smooth muscle. You know from physiology that it has two main switches. click The first switch closes the airway. The vagus nerve releases acetylcholine, which acts on M3 muscarinic receptors. Through Gq and IP3, calcium rises in the cell, and the muscle contracts. Muscarinic antagonists block this receptor. Ipratropium is short-acting, and tiotropium is long-acting, so we call it a LAMA. click The second switch opens the airway. Beta-2 receptors use Gs to make cyclic AMP, and cyclic AMP relaxes the muscle. Beta-2 agonists stimulate this receptor. Salbutamol, also called albuterol, is short-acting, so it is a SABA. Formoterol and salmeterol are long-acting, so they are LABAs. click There are two more targets. Phosphodiesterase breaks down cyclic AMP, and theophylline blocks it. And leukotrienes from mast cells also contract the muscle, so montelukast blocks their receptor.

Say carefully: acetylcholine = uh-SEE-tul-KOH-leen; **muscarinic** = mus-kuh-RIN-ik; **antagonist** = an-TAG-uh-nist; **ipratropium** = ip-ruh-TROH-pee-um; **tiotropium** = ty-oh-TROH-pee-um; **LAMA** = LAH-muh; **agonist** = AG-uh-nist; **salbutamol** = sal-BYOO-tuh-mol; **albuterol** = al-BYOO-tuh-rol; **SABA** = SAH-buh; **formoterol** = for-MOH-tuh-rol; **salmeterol** = sal-MET-uh-rol; **LABA** = LAH-buh; **phosphodiesterase** = FOS-foh-dy-ES-tuh-rayz; **theophylline** = thee-OF-uh-lin; **montelukast** = mon-tuh-LOO-kast

Time: about 80 seconds, 3 clicks

Slide 20: Side Effects Explained by Receptors

What to say: Here is a useful idea: if you know the receptor, you can predict the side effect. Beta-2 receptors are also in skeletal muscle, so beta-2 agonists cause tremor. They also push potassium into cells, so they can cause hypokalemia, and at high doses, tachycardia. M3 receptors are also in the salivary glands, so muscarinic antagonists cause a dry mouth. It also works the other way. Propranolol is a non-selective beta-blocker. It blocks the beta-2 receptors in the airway, so in a patient with asthma it can cause severe bronchospasm. Finally, inhaled steroids land in the mouth and throat and lower the local immunity there, so they can cause oral thrush. That is why patients should rinse their mouth after each dose.

Say carefully: skeletal = SKEL-uh-tul; **tremor** = TREM-er; **potassium** = puh-TAS-ee-um; **hypokalemia** = HY-poh-kuh-LEE-mee-uh; **tachycardia** = tak-ih-KAR-dee-uh; **salivary** = SAL-uh-vair-ee; **propranolol** = proh-PRAN-uh-lol; **bronchospasm** = BRONG-koh-spaz-um; **thrush** = THRUSH

Time: about 65 seconds, no clicks

Slide 21: Corticosteroids and Biologics

What to say: Bronchodilators treat the muscle. But remember the late phase: we also need to treat the inflammation. click Corticosteroids enter the cell and bind the glucocorticoid receptor. The complex moves into the nucleus and switches off inflammatory genes, for example through NF-kappa-B. So there are fewer cytokines, fewer eosinophils and less mucus. Steroids also increase the number of beta-2 receptors, so they work well together with a LABA. click For severe asthma, we also have biologics. These are antibodies that block the Th2 pathway from part one. Omalizumab binds IgE, mepolizumab blocks interleukin five, and dupilumab blocks the receptor for interleukins four and thirteen. click In COPD, the inflammation is mostly neutrophilic, so steroids help less. They help mainly patients with frequent flare-ups and high blood eosinophils.

Say carefully: glucocorticoid = gloo-koh-KOR-tih-koyd; **NF-κB** = en-eff-KAP-uh-bee; **cytokine** = SY-tuh-kyn; **biologic** = by-uh-LOJ-ik; **omalizumab** = oh-muh-LIZ-yoo-mab; **mepolizumab** = mep-uh-LIZ-yoo-mab; **dupilumab** = doo-PIL-yoo-mab; **neutrophilic** = noo-truh-FIL-ik

Time: about 70 seconds, 3 clicks

Slide 22: Treatment: Asthma vs COPD

What to say: Now we can match each drug to each disease. click In asthma, the main problem is inflammation. So a SABA alone is not enough: it relaxes the muscle but leaves the inflammation untreated, and SABA-only treatment is linked to severe attacks and deaths. Every patient needs an inhaled corticosteroid, or ICS. The preferred reliever is ICS-formoterol, because formoterol works within minutes and lasts twelve hours. In mild asthma, patients use it only when needed; in more severe asthma, they use it every day as well. click In COPD, the main problem is air trapping. A LAMA plus a LABA keeps the airways open longer, so the lungs empty better and the patient is less breathless. We add an ICS only if there are flare-ups and high eosinophils, and if flare-ups continue, roflumilast or azithromycin can help. And every patient needs to stop smoking, get vaccinated and join pulmonary rehabilitation.

Say carefully: ICS = eye-see-ES; **reliever** = rih-LEE-ver; **MART (on the slide)** = MART; **roflumilast** = roh-FLOO-mih-last; **azithromycin** = ay-ZITH-roh-MY-sin; **vaccinated** = VAK-sih-nay-tid; **rehabilitation** = REE-huh-bil-ih-TAY-shun

Time: about 80 seconds, 2 clicks

Slide 23: Smoking Cessation and Oxygen Therapy

What to say: Two treatments change the course of COPD. The first is stopping smoking. This classic graph shows FEV1 over a lifetime. Everyone loses some lung function with age, but a susceptible smoker loses it much faster. If he stops, the lost function does not come back, but the decline slows to the normal rate, even at sixty-five. So it is never too late to quit. click The second is long-term oxygen for patients with severe hypoxemia at rest. It relieves hypoxic vasoconstriction and improves survival. But be careful: some patients retain carbon dioxide. If we give them too much oxygen, the vasoconstriction in poorly ventilated areas is released, the V/Q mismatch gets worse, and their carbon dioxide rises. That is why we aim for an oxygen saturation of eighty-eight to ninety-two percent.

Say carefully: susceptible = suh-SEP-tuh-bul; **decline** = dih-KLYN; **survival** = ser-VY-vul; **V/Q** = vee-over-KYOO; **saturation** = sach-uh-RAY-shun

Time: about 70 seconds, 1 click

Slide 24: Quick Quiz: Asthma or COPD?

What to say: Now it's your turn. I will read four sentences. Please call out the answer: asthma or COPD? Number one: Th2 cells and eosinophils drive the inflammation. click Yes, asthma. Number two: elastase destroys the alveolar walls. click Correct, COPD. Number three: spirometry becomes normal after a bronchodilator. click Asthma, because it is reversible. Number four: a low DLCO and a barrel chest. click COPD, emphysema. Well done, everyone!

Time: about 50 seconds, 4 clicks (includes pauses for the audience)

Slide 25: Back to Our Patients

What to say: So, let's go back to our question. Why does Sara's obstruction come and go, while Mr. Brown's stays? Sara's obstruction comes from Type 2 inflammation and overreactive muscle. These change from day to day, and they reverse with treatment. Her DLCO is normal, and she needs an anti-inflammatory reliever: ICS-formoterol. Mr. Brown's obstruction comes from structural damage: elastase destroyed his alveoli, and mucus and fibrosis narrowed his small airways. This damage cannot be reversed. He must stop smoking, and he needs a LAMA plus a LABA. click So our take-home message is this: the same symptom can come from two very different mechanisms. When you meet a breathless patient, ask: does the obstruction come and go, or does it stay?

Say carefully: structural = STRUK-cher-ul; **anti-inflammatory** = AN-tee-in-FLAM-uh-tor-ee; **mechanism** = MEK-uh-niz-um

Time: about 65 seconds, 1 click

Slide 26: Thank You for Your Attention!

What to say: That brings us to the end of our presentation. Thank you very much for listening. We are happy to answer your questions.

Time: about 10 seconds, no clicks

Questions you might get (Presenter 3)

1. **Q:** Why is formoterol, and not salmeterol, used as a reliever?
   A: Formoterol starts working within a few minutes, like a SABA, and it lasts about twelve hours. Salmeterol starts more slowly, so it is not suitable as a reliever.
2. **Q:** Why don’t antihistamines work well in asthma?
   A: Histamine is only one of many mediators. Leukotrienes and the late-phase inflammation are more important, so blocking histamine alone has very little effect.
3. **Q:** Why can high-flow oxygen be dangerous in some COPD patients?
   A: In patients who retain CO2, high oxygen releases hypoxic vasoconstriction in poorly ventilated areas, so the V/Q mismatch gets worse. Oxygen also releases CO2 from hemoglobin (the Haldane effect). Their CO2 rises, which can cause acidosis and drowsiness, so we aim for a saturation of 88 to 92 percent.

References (slide 27)

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